Multimedia Wireless Networks A2

ns-3 Setup & First Scenario

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Basic Path

Goal



Does EDCA prioritization still work with realistic multimedia traffic?

ns-3 setup

```
❯ ./ns3 build
[0/2] Re-checking globbed directories...
[1885/2635] Building CXX object contrib/netsimulyzer/examples/CMakeFiles/wifi-bianchi-netsimulyzer.dir/wifi-bianchi-net
simulyzer.cc.o
/home/sascha/ns-allinone-3.48/ns-3.48/contrib/netsimulyzer/examples/wifi-bianchi-netsimulyzer.cc: In function ‘int main
(int, char**)’:
/home/sascha/ns-allinone-3.48/ns-3.48/contrib/netsimulyzer/examples/wifi-bianchi-netsimulyzer.cc:3417:22: warning: vari
able ‘k’ set but not used [-Wunused-but-set-variable=]
 3417 |             uint32_t k = 0;
      |                      ^
[2634/2635] Linking CXX executable /home/sascha/ns-allinone-3.48/ns-3.48/build/utils/ns3.48-test-runner-default
Finished executing the following commands:
/usr/bin/cmake --build /home/sascha/ns-allinone-3.48/ns-3.48/cmake-cache -j 7
```

Item	Value
ns-3 version	ns-3.48
Operating system	Fedora Linux 44
Kernel	7.2.7-200.fc44.x86_64 (64 bit)
Desktop	KDE Plasma 6.7.5 (Frameworks 6.30.0, Qt 6.11.2), Wayland
Device	Lenovo ThinkPad X1 Carbon Gen 9 (20XXS33900)
CPU	Intel Core i7-1165G7 @ 2.80 GHz, 8 threads
Memory	32 GiB (31.1 GiB usable)
GPU	Intel Iris Xe Graphics

```
❯ ./test.py
[0/2] Re-checking globbed directories...
ninja: no work to do.
Finished executing the following commands:
/usr/bin/cmake --build /home/sascha/ns-allinone-3.48/ns-3.48/cmake-cache -j 7
```

```
989 of 989 tests passed (989 passed, 0 skipped, 0 failed, 0 crashed, 0 valgrind errors)
```

3GPP TR 26.926 V19.0.0 (Rel. 19)

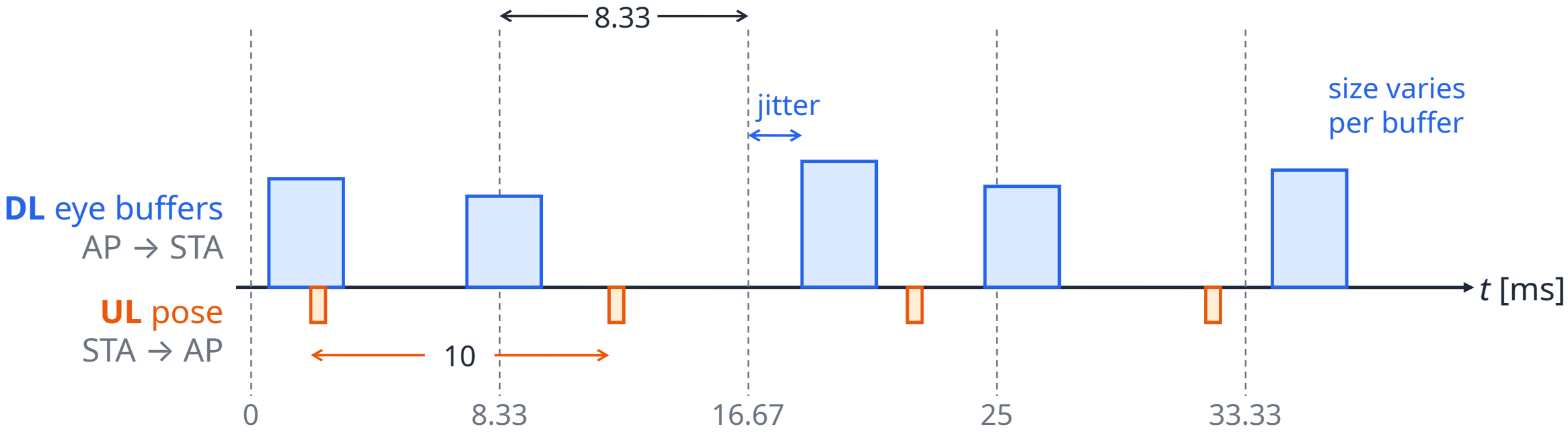
TS
Technical Specification

TR
Technical Report

Table 6.5.3.1-1: Statistical parameter values for dual eye buffer frame size (TR 38.838)

Parameter	unit	values for evaluation	Optional values for evaluation
Mean: M	byte	R×1e6 / (2×F) /8	R×1e6 / (2×F) / 8
STD	byte	10.5% of M	4% of M
Max	byte	150% of M	112% of M
Min	byte	50% of M	88% of M
R: data rate of the flow in Mbps F: frame generation rate of the flow in fps			

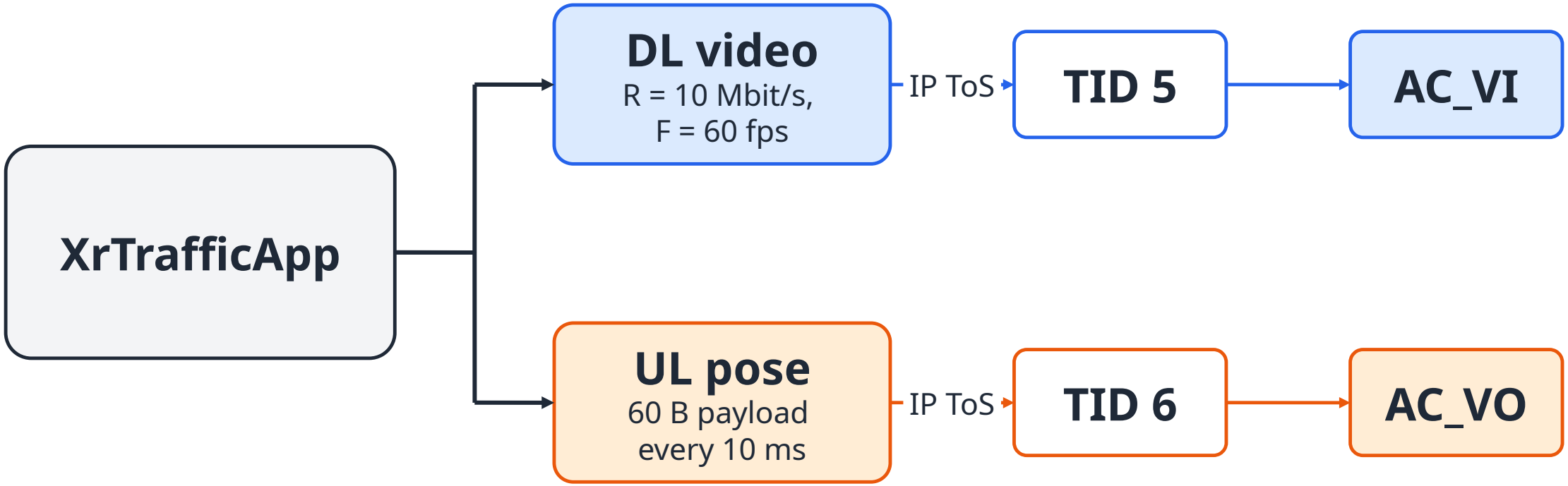
3GPP Traffic Model



Frame size: truncated Gaussian, $M = R \cdot 10^6 / (2F) / 8$, STD 10.5 % of M, ±50 %
Jitter: truncated Gaussian, STD 2 ms, ±4 ms

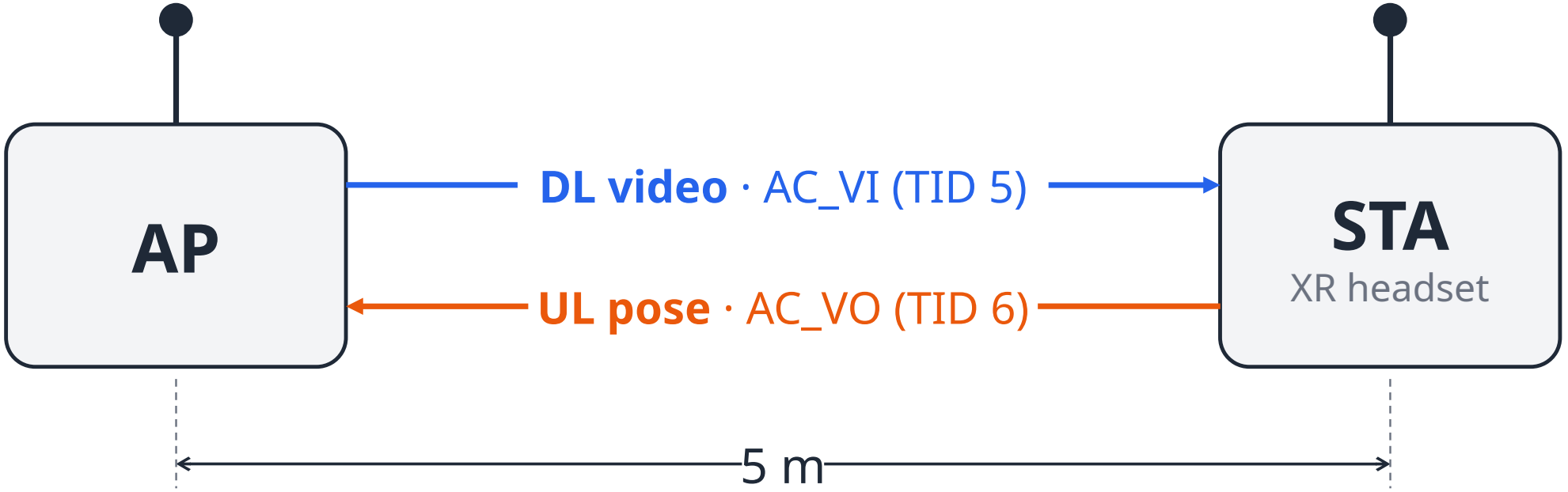
Based on Table 6.5.3.1-1

Traffic Generator



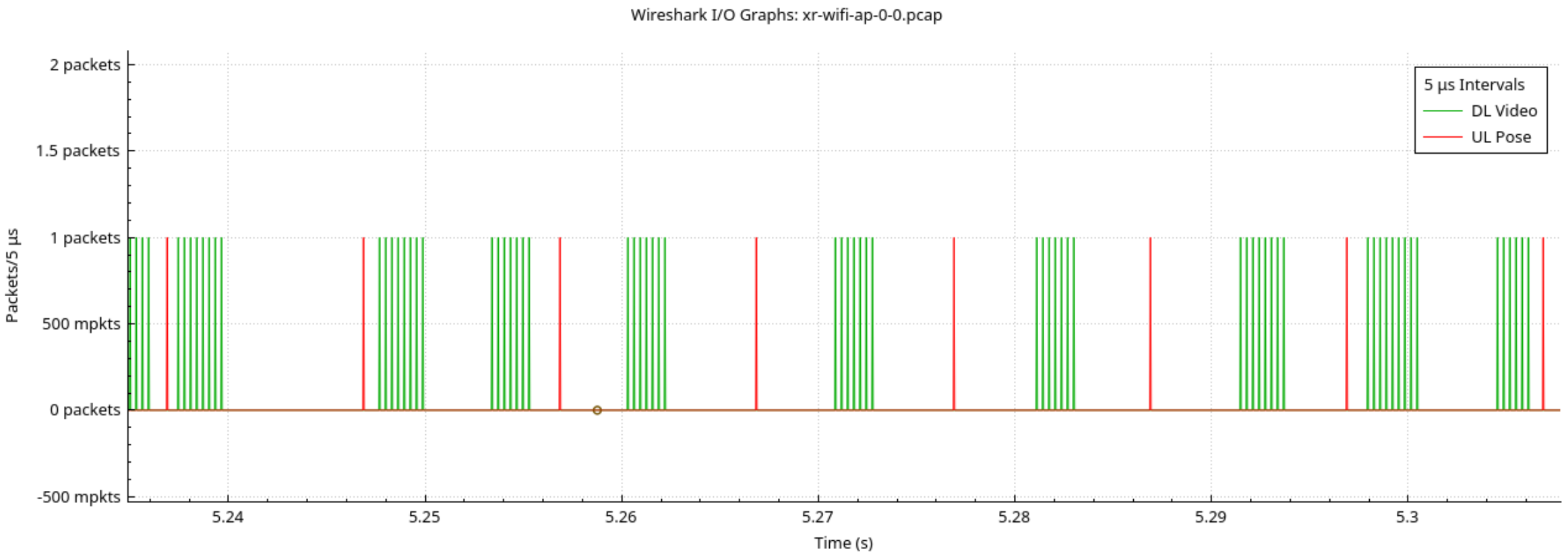
ToS byte → 802.11 user priority (TID) → access category

Scenario

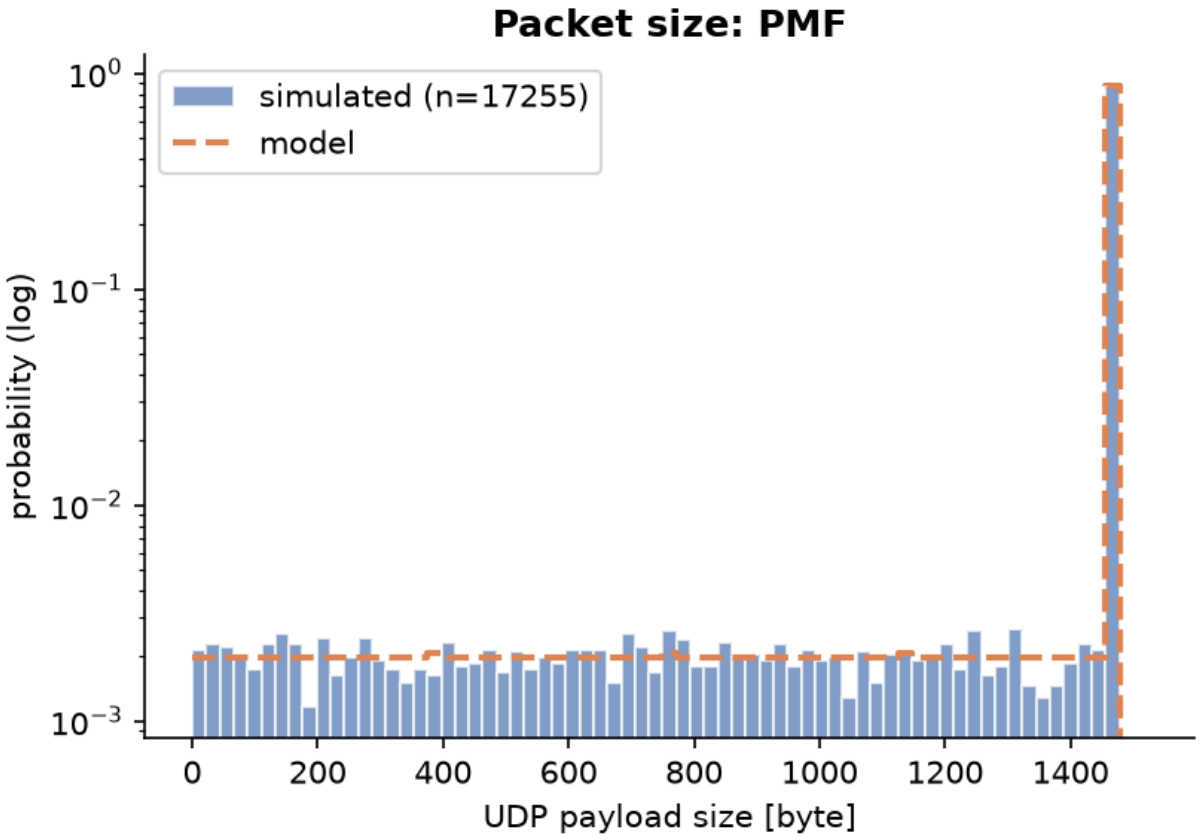


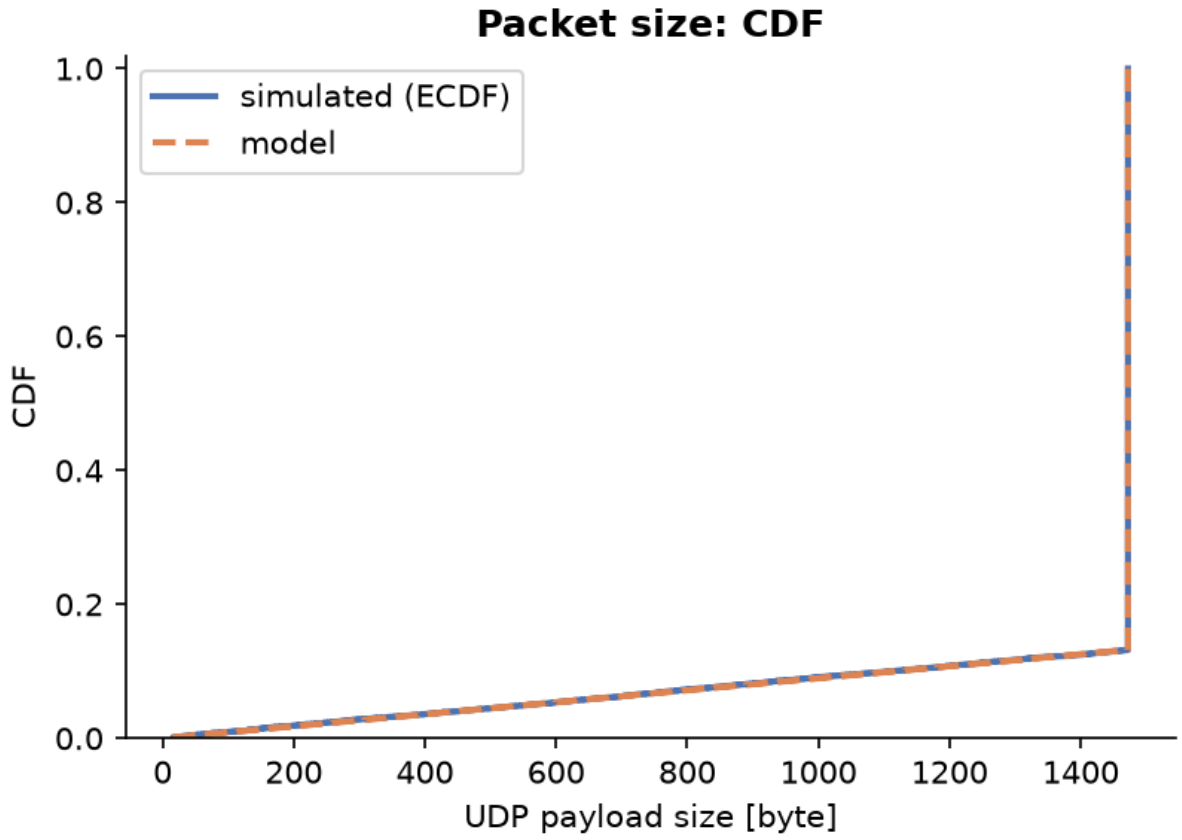
802.11a · 54 Mbit/s · channel 36

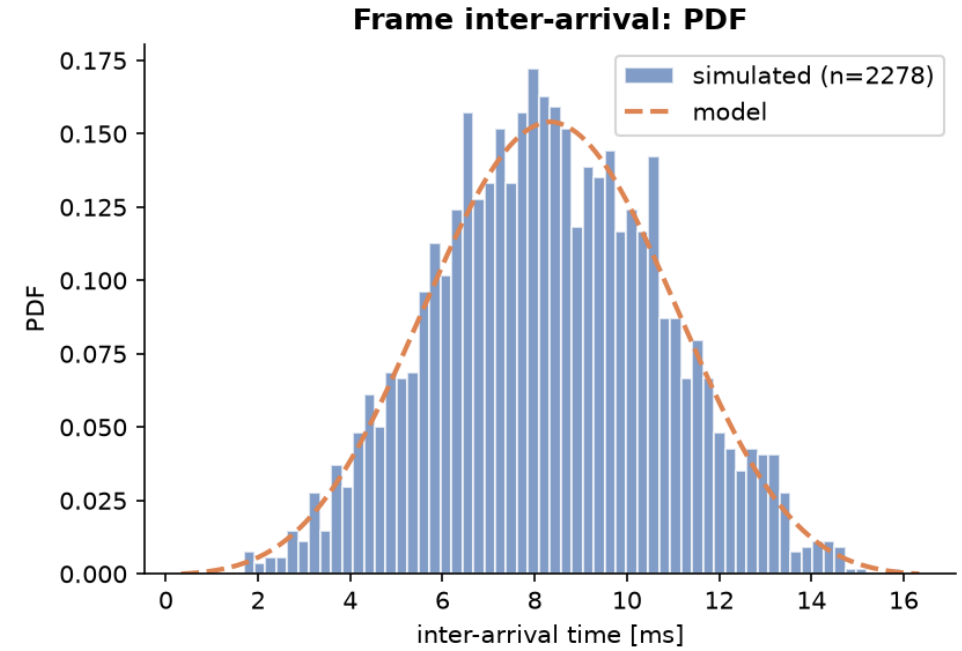
Verification

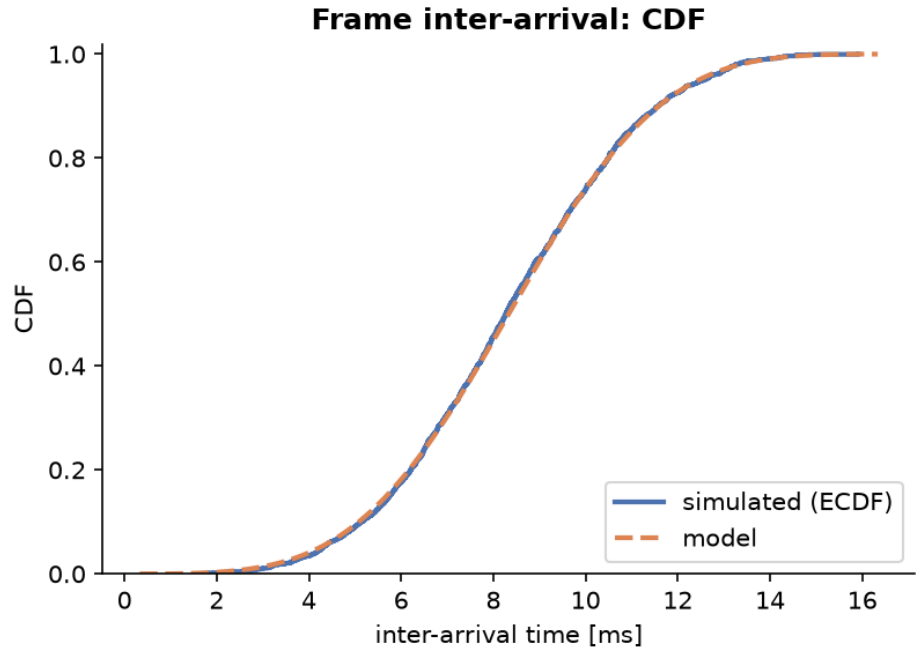


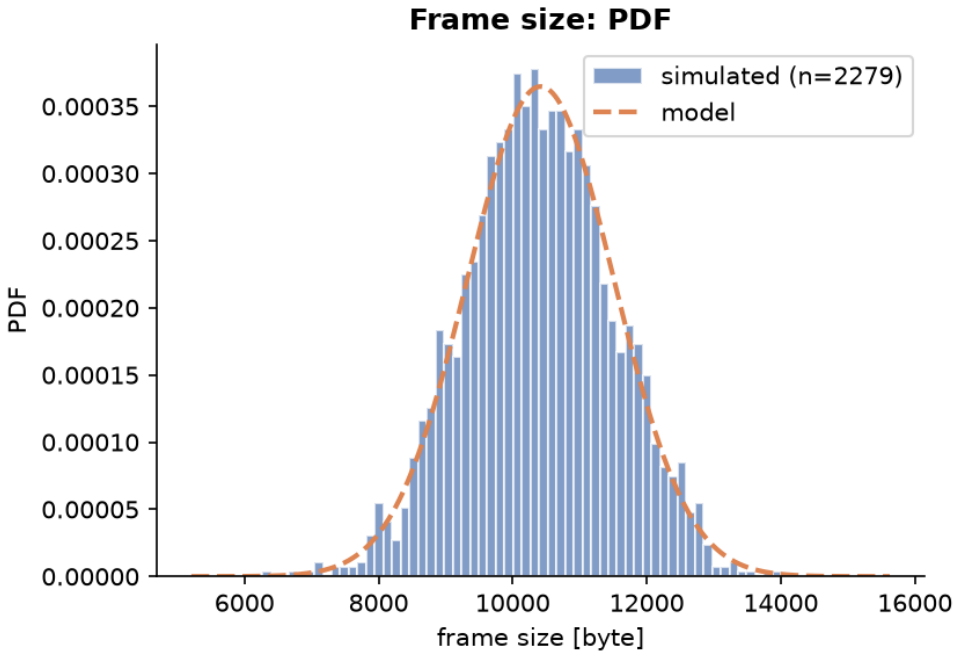
Verification

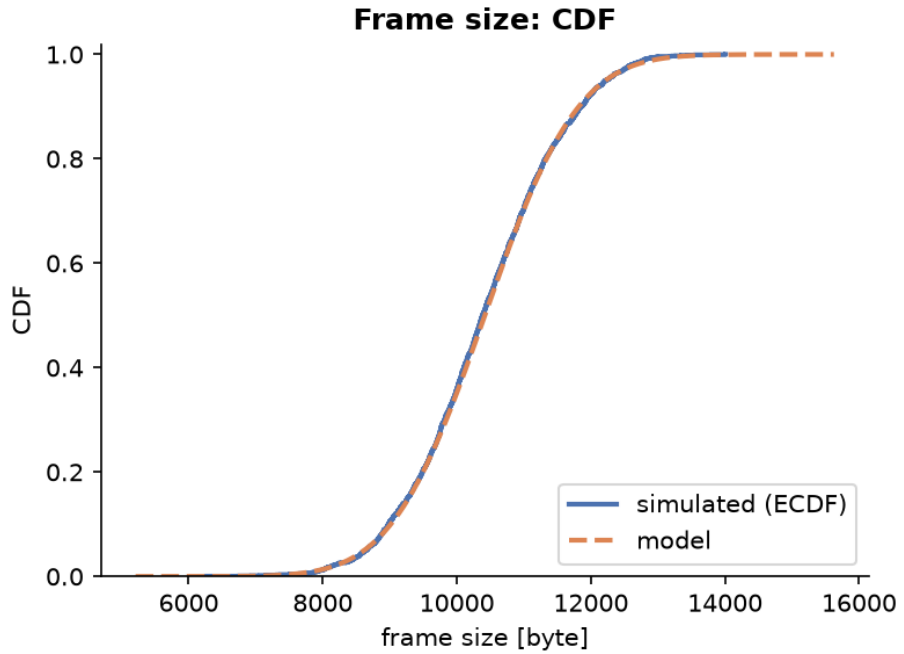





Frame inter-arrival: PDF
simulated (n=2278)
model
PDF
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
0
2
4
6
8
10
12
14
16
inter-arrival time [ms]

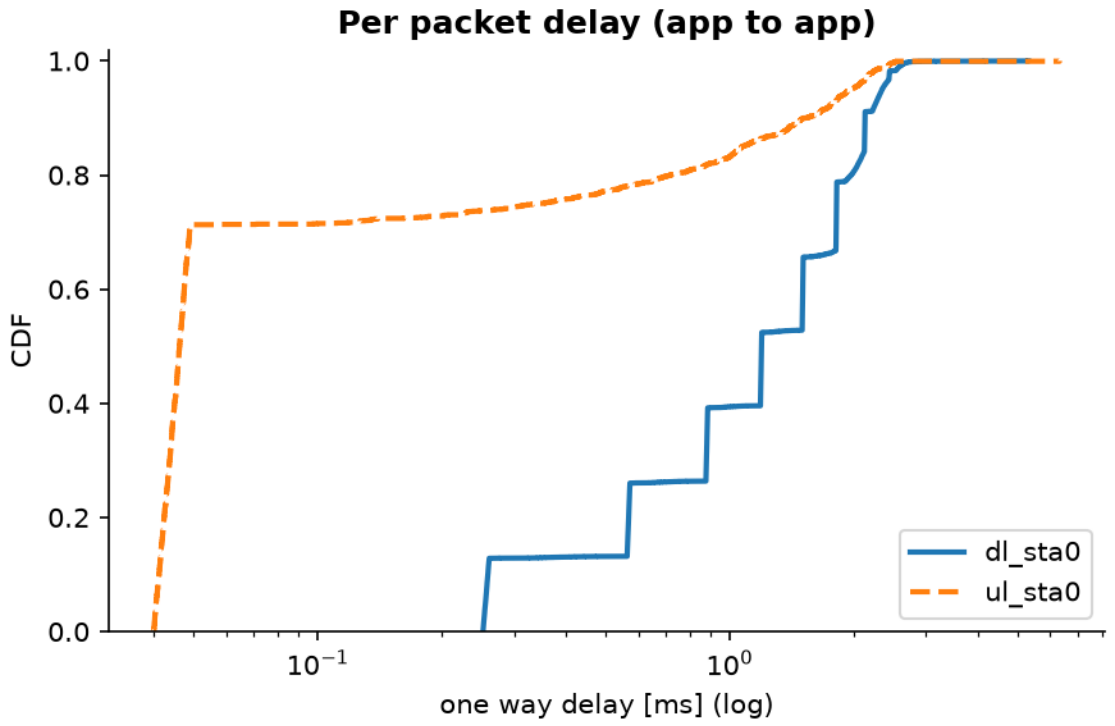

Frame inter-arrival: CDF
simulated (ECDF)
model
CDF
1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
10
12
14
16
inter-arrival time [ms]


Frame size: PDF
simulated (n=2279)
model
PDF
0.00035
0.00030
0.00025
0.00020
0.00015
0.00010
0.00005
0.00000
6000
8000
10000
12000
14000
16000
frame size [byte]


Frame size: CDF
simulated (ECDF)
model
CDF
1.0
0.8
0.6
0.4
0.2
0.0
6000
8000
10000
12000
14000
16000
frame size [byte]

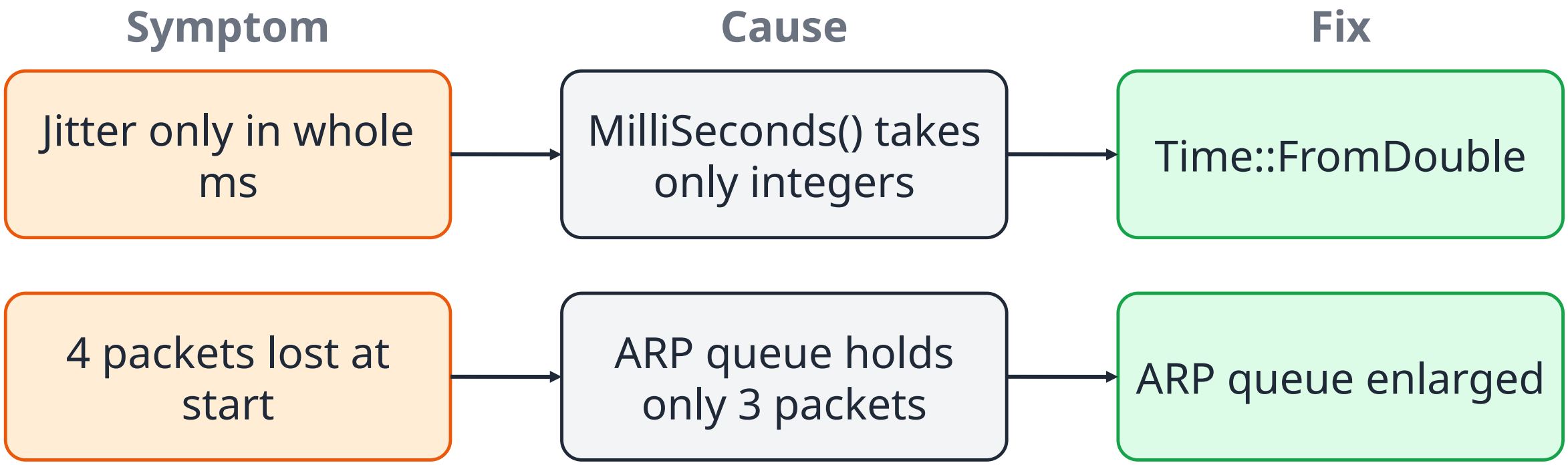
Results

	DL video	UL pose
throughput	10.2 Mbit/s	0.07 Mbit/s
mean delay	1.29 ms	0.37 ms




```
No.  Time  Source  Destination  Protoc  Length  Info
> Frame 100: Packet, 1560 bytes on wire (12480 bits), 1560 bytes c
> Radiotap Header v0, Length 22
v 802.11 radio information
    PHY type: 802.11a (OFDM) (5)
    Turbo type: Non-turbo (0)
    Data rate: 54.0 Mb/s
    Channel: 36
    Frequency: 5180MHz
    TSF timestamp: 1041846
  > [Duration: 252µs]
v IEEE 802.11 QoS Data, Flags: ......F.C
    Type/Subtype: QoS Data (0x0028)
  v Frame Control Field: 0x8802
      .... ..00 = Version: 0
      .... 10.. = Type: Data frame (2)
      1000 .... = Subtype: 8
    > Flags: 0x02
    .000 1101 1100 1100 = Duration: 3532 microseconds
  > Receiver address: 00:00:00_00:00:01 (00:00:00:00:00:01)
  > Transmitter address: 00:00:00_00:00:02 (00:00:00:00:00:02)
  > Destination address: 00:00:00_00:00:01 (00:00:00:00:00:01)
  > Source address: 00:00:00_00:00:02 (00:00:00:00:00:02)
  > BSS Id: 00:00:00_00:00:02 (00:00:00:00:00:02)
  > STA address: 00:00:00_00:00:01 (00:00:00:00:00:01)
    .... .... .... 0000 = Fragment number: 0
    0000 0001 1110 .... = Sequence number: 30
    Frame check sequence: 0x00000000 [unverified]
    [FCS Status: Unverified]
    [WLAN Flags: ......F.C]
  v Qos Control: 0x0005
      .... .... .... 0101 = TID: 5
      [.... .... .... .101 = Priority: Video (Video) (5)]
      .... .... ...0 .... = EOSP: Service period
      .... .... .00. .... = Ack Policy: Normal Ack (0x0)
      .... .... 0... .... = Payload Type: MSDU
    > 0000 0000 .... .... = QAP PS Buffer State: 0x00
```

What broke


Symptom
Cause
Fix
Jitter only in whole ms
MilliSeconds() takes only integers
Time::FromDouble
4 packets lost at start
ARP queue holds only 3 packets
ARP queue enlarged

Recap

